Performance Evaluation Metrics in Classification

A Complete Guide with Examples

For Action Recognition Using V-JEPA 2 and Qwen2.5-VL

This document explains **every evaluation metric** used in the thesis:

Top-1 Accuracy · Top-5 Accuracy · Precision · Recall
F1-Score · MCC · Cohen's Kappa · Confusion Matrix

Thesis Reference Document

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Before We Begin: The Four Fundamental Counts

Before understanding any metric, you need to understand four basic concepts that **every metric is built from**. These come from the **confusion matrix**.

The Binary Case (Two Classes)

Imagine a simple scenario: a model must decide if a video shows "**Vault**" or "**Not Vault**".

The Four Outcomes

When a model makes a prediction, exactly one of four things happens:

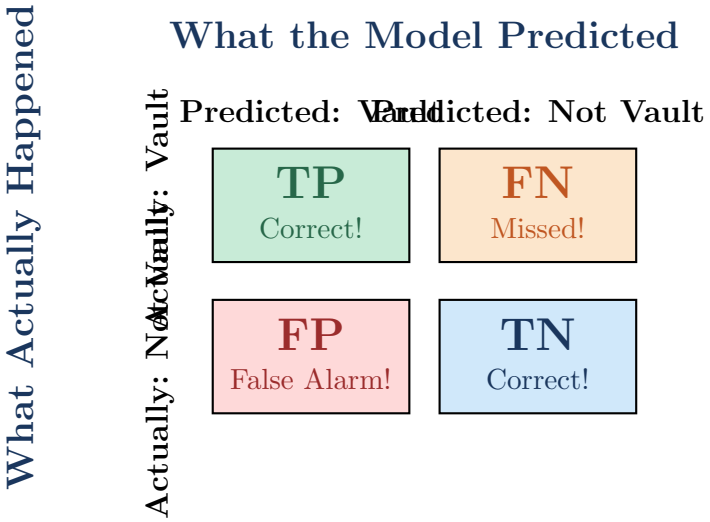
TP **True Positive:** The video *is* Vault, and the model *correctly says* "Vault."
⇒ The model got it right on a positive example.

TN **True Negative:** The video is *not* Vault, and the model *correctly says* "Not Vault."
⇒ The model got it right on a negative example.

FP **False Positive:** The video is *not* Vault, but the model *wrongly says* "Vault."
⇒ False alarm! The model cried wolf.

FN **False Negative:** The video *is* Vault, but the model *wrongly says* "Not Vault."
⇒ The model missed a real case!

Visual Representation: The Confusion Matrix



Multi-Class Extension

In your thesis, you have **4 classes** (FineGym) or **51 classes** (HMDB51). For multi-class problems, we compute TP, FP, FN *for each class separately* using a "one-vs-rest" approach:

FineGym Example: Computing TP / FP / FN for Balance Beam

For the Balance Beam class specifically:

- **TP** = Videos that ARE Balance Beam and model said "Balance Beam"
- **FP** = Videos that are NOT Balance Beam but model wrongly said "Balance Beam" (maybe they were Floor Exercise)
- **FN** = Videos that ARE Balance Beam but model said something else (maybe "Uneven Bars")

Then we repeat this for Floor Exercise, Uneven Bars, and Vault.

Top-1 Accuracy

The Simple Question

What Top-1 Accuracy Asks

"Is the model's single best guess correct?"

The model looks at a video and outputs probabilities for every class. We take the class with the **highest probability** (the model's "best guess") and check: does it match the true label?

The Formula

Mathematical Definition

$$\text{Top-1 Accuracy} = \frac{\text{Number of correctly classified samples}}{\text{Total number of samples}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\hat{y}_i = y_i\right]$$

where:

- N = total number of test samples
- y_i = the true label of sample i (ground truth)
- $\hat{y}_i = \arg\max_c \, p(c \mid \mathbf{x}_i)$ = the class with the highest predicted probability
- $\mathbf{1}[\cdot]$ = indicator function (equals 1 if the condition is true, 0 otherwise)

Worked Example

Step-by-Step Example with 10 Videos

Suppose we test on 10 FineGym videos. The model's top-1 prediction for each:

Video	True Label	Model's Top-1 Prediction	Correct?
1	Balance Beam	Balance Beam	✓
2	Floor Exercise	Floor Exercise	✓
3	Uneven Bars	Balance Beam	✗
4	Vault	Vault	✓
5	Balance Beam	Balance Beam	✓
6	Floor Exercise	Vault	✗
7	Uneven Bars	Uneven Bars	✓
8	Vault	Vault	✓
9	Balance Beam	Balance Beam	✓
10	Floor Exercise	Floor Exercise	✓

$$\text{Top-1 Accuracy} = \frac{8 \text{ correct}}{10 \text{ total}} = 0.80 = 80\%$$

Why Top-1 Accuracy Is Needed

1. **Most intuitive metric** — everyone understands "80% of predictions were correct"
2. **Universal benchmark** — every paper in action recognition reports it, enabling direct comparison
3. **Strictest test** — the model must get its single best guess exactly right

The Dangerous Limitation of Accuracy

Why Accuracy Alone Can Be Misleading

The Class Imbalance Problem:

Imagine a dataset with 950 "walking" videos and 50 "falling" videos (total = 1000). A completely *useless* model that **always predicts "walking"** (ignoring the input entirely) would get:

$$\text{Accuracy} = \frac{950}{1000} = 95\%$$

That looks impressive! But the model **never detects a single fall**. It has completely failed at the important task.

This is exactly why we need additional metrics like F1, MCC, and Kappa.

Top-5 Accuracy

The Softer Question

What Top-5 Accuracy Asks

"Is the correct answer anywhere in the model's top 5 guesses?"

Instead of requiring the #1 prediction to be correct, we check if the true label appears in the 5 most confident predictions. This gives the model some "room for error."

The Formula

Mathematical Definition

$$\text{Top-5 Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[y_i \in \text{Top5}(\hat{\mathbf{p}}_i)\right]$$

where $\text{Top5}(\hat{\mathbf{p}}_i)$ returns the 5 classes with the highest predicted probabilities for sample i.

Worked Example

Step-by-Step: One HMDB51 Video

The model sees a video of someone **fencing** and outputs probabilities for all 51 classes. The top 5 are:

Rank	Predicted Class	Probability
1	Sword Exercise	0.25
2	Kick	0.18
3	**Fencing**	**0.14**
4	Punch	0.12
5	Draw Sword	0.09
6–51	(other 46 classes)	(remaining 0.22)

The true label is **Fencing**:

- **Top-1:** ✗ Incorrect — the #1 prediction is "Sword Exercise," not "Fencing"
- **Top-5:** ✓Correct — "Fencing" appears at rank #3, which is within the top 5

Why Top-5 Accuracy Only for HMDB51, Not FineGym

Important: Top-5 Is Meaningless for 4 Classes

FineGym has only 4 classes (BB, FX, UB, VT).

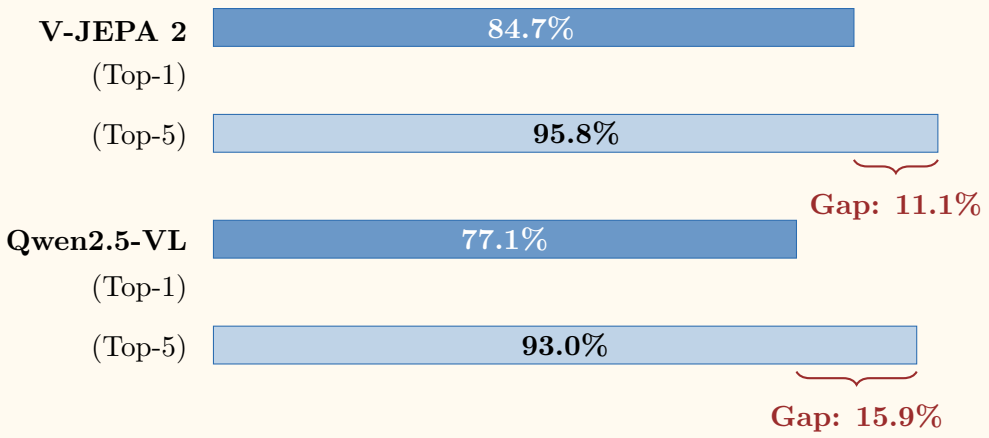
If you check whether the correct class is in the "top 5" of 4 classes, the answer is **always yes** — because all 4 classes *are* within the top 5. The metric would be 100% for every model, telling you nothing.

HMDB51 has 51 classes, so Top-5 is meaningful — the correct class genuinely might not be in the top 5 guesses.

What the Top-1 vs Top-5 Gap Tells You

Reading the Gap Between Top-1 and Top-5

The difference between Top-1 and Top-5 accuracy reveals **how confused the model is between similar classes**.



Interpretation:

- **V-JEPA 2** (gap = 11.1%): The model is fairly confident. When it's wrong, the correct answer is usually its #2 or #3 guess.
- **Qwen2.5-VL** (gap = 15.9%): More confusion between similar classes. It knows the "neighborhood" but struggles with the exact class more often.

Smaller gap = more precise model. Larger gap = model is confused between similar classes.

Precision

The Core Question

What Precision Asks

"When the model says 'this is class X,' how often is it actually correct?"

Precision measures **how trustworthy** the model's predictions are. High precision means: when the model says "Vault," you can trust that it really is Vault.

The Formula

Mathematical Definition

For a single class c:

$$\boxed{\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}}$$

In words:

$$\text{Precision} = \frac{\text{Correctly predicted as class } c}{\text{Everything the model predicted as class } c}$$

Worked Example

FineGym: Precision for "Balance Beam"

Suppose out of 55 test videos total:

- The model predicted 18 videos as "Balance Beam"
- Of those 18 predictions, **15 were actually Balance Beam** (TP = 15)
- But **3 were actually Uneven Bars** that the model got wrong (FP = 3)

$$\text{Precision}_{\text{BB}} = \frac{15}{15+3} = \frac{15}{18} = 0.833 = 83.3\%$$

Meaning: When the model says "Balance Beam," it's right 83.3% of the time.

Macro vs. Weighted Precision

Since we have multiple classes, we need to combine per-class precisions into a single number.

Macro Precision

$$\frac{1}{C}\sum_{c=1}^{C}\text{Precision}_c$$

Simple average of all per-class precisions.

Every class counts **equally**, whether it has 10 or 1000 samples.

Use when: All classes matter equally.

Weighted Precision

$$\sum_{c=1}^{C}\frac{n_c}{N}\cdot\text{Precision}_c$$

Weighted average where larger classes have more influence.

A class with 200 samples matters more than one with 20.

Use when: Reflecting overall volume.

Macro vs Weighted — Numerical Example

Suppose 3 classes with different numbers of test samples:

Class	Precision	Test Samples	Weight (n_c/N)
Balance Beam	90%	100	$100/200 = 0.50$
Floor Exercise	80%	80	$80/200 = 0.40$
Vault	60%	20	$20/200 = 0.10$
Total		200	1.00

Macro Precision $= \frac{90+80+60}{3} =$ **76.7%**

Weighted Precision $= 0.50 \times 90 + 0.40 \times 80 + 0.10 \times 60 = 45 + 32 + 6 =$ **83.0%**

Weighted is higher because the model does best on Balance Beam, which has the most samples.

Recall (Sensitivity)

The Core Question

What Recall Asks

"Of all the videos that actually ARE class X, how many did the model successfully find?"

Recall measures **how completely** the model detects all instances of a class. High recall means: the model rarely *misses* a real instance.

The Formula

Mathematical Definition

For a single class c:

$$\boxed{\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}}$$

In words:

$$\text{Recall} = \frac{\text{Correctly identified instances of class } c}{\text{All actual instances of class } c}$$

Worked Example

FineGym: Recall for "Balance Beam"

Suppose there are **20 actual Balance Beam videos** in the test set:

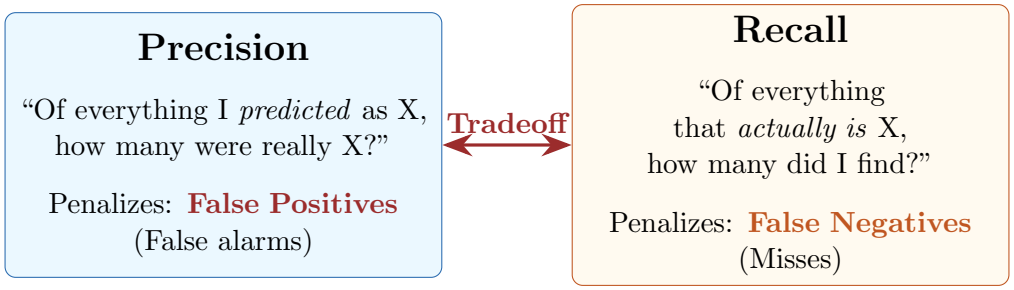
- The model correctly identified **15 of them** as Balance Beam (TP = 15)
- But it **missed 5** — it wrongly called them Uneven Bars or Floor Exercise (FN = 5)

$$\text{Recall}_{\text{BB}} = \frac{15}{15 + 5} = \frac{15}{20} = 0.75 = 75\%$$

Meaning: The model found 75% of all actual Balance Beam videos. It missed 25%.

Precision vs. Recall: The Fundamental Tradeoff

These two metrics are **inherently in tension**:



Real-World Analogy

Think of a **security guard** checking IDs at a gate:

- **High Precision, Low Recall:** The guard is very strict — only lets people through when absolutely certain. Almost never lets in an intruder (low FP), but frequently turns away legitimate visitors (high FN).
- **High Recall, Low Precision:** The guard lets almost everyone in to make sure no legitimate visitor is rejected. Catches all legitimate visitors (low FN), but also lets in some intruders (high FP).
- **High Both:** A perfect guard who correctly identifies almost everyone.

F1-Score

Why We Need F1

Since precision and recall are in tension, we want a **single number** that balances both. That's the F1-Score.

What F1-Score Is

The harmonic mean of Precision and Recall.

It gives a single number that is high **only when BOTH precision and recall are high**. If either one is low, F1 drops dramatically.

The Formula

Mathematical Definition

$$F1_c = 2 \times \frac{\text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} = \frac{2 \cdot TP_c}{2 \cdot TP_c + FP_c + FN_c}$$

Why Harmonic Mean, Not Regular Average?

This is a critical point. The harmonic mean **punishes imbalance** much more harshly:

Precision	Recall	Regular Average	Harmonic Mean (F1)
100%	0%	50%	**0%**
90%	10%	50%	**18%**
80%	40%	60%	**53%**
80%	80%	80%	**80%**
95%	90%	92.5%	**92.4%**

Why This Matters

A model that has 100% precision but 0% recall (it never predicts any class, so it's never wrong but also never finds anything) would get 50% on a regular average — which sounds decent. But F1 correctly gives it **0%**, reflecting that the model is useless.

Worked Example

Complete F1 Calculation for "Balance Beam"

From our earlier examples:

- $\text{Precision}_{\text{BB}} = 83.3\%$ (15 correct out of 18 predictions)
- $\text{Recall}_{\text{BB}} = 75.0\%$ (15 found out of 20 actual)

$$F1_{\text{BB}} = 2 \times \frac{0.833 \times 0.750}{0.833 + 0.750} = 2 \times \frac{0.625}{1.583} = 2 \times 0.395 = \mathbf{0.789 = 78.9\%}$$

Macro F1 vs. Weighted F1

Variant	Formula	Used For
Macro F1	$\frac{1}{C}\sum_{c=1}^{C} F1_c$	Model selection in your code
Weighted F1	$\sum_{c=1}^{C} \frac{n_c}{N} F1_c$	Overall performance summary

F1 Is Your Primary Model Selection Metric

In your code, the best classifier is chosen by **Macro F1**, not accuracy:

```
if f1 > best_f1:  best_f1 = f1; best_name = cname
```

This is the correct choice because Macro F1 is robust to class imbalance and ensures rare classes are not ignored.

Matthews Correlation Coefficient (MCC)

The Most Balanced Metric

What MCC Is

MCC is a **correlation coefficient** between the model's predictions and the actual ground truth. It is widely regarded as the **single most informative and balanced metric** for classification.

It is the **only metric that uses all four cells** of the confusion matrix (TP, TN, FP, FN) symmetrically.

The Formula

Mathematical Definition (Binary Case)

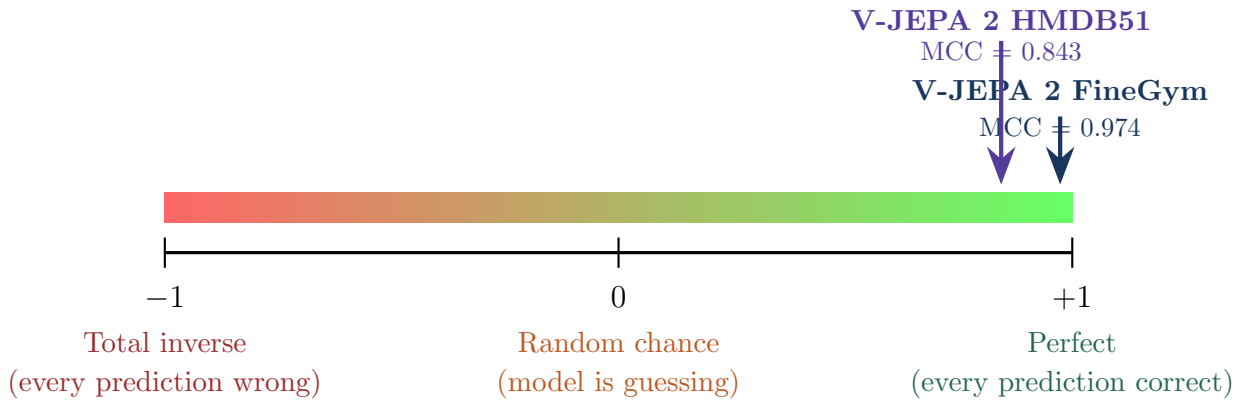
$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

For multi-class (what scikit-learn computes), it generalizes using the full confusion matrix $\mathbf{C}$:

$$MCC = \frac{c \cdot s - \sum_k p_k \cdot t_k}{\sqrt{(s^2 - \sum_k p_k^2)(s^2 - \sum_k t_k^2)}}$$

where c = total correct predictions, s = total samples, p_k = times class k was predicted, t_k = true occurrences of class k.

Value Range and Interpretation



Why MCC Is Special — The "Accuracy Trap" Example

MCC Catches What Accuracy Misses

Scenario: 950 test samples are class A, 50 are class B.
Useless model: Always predicts class A (ignores input completely).

	Accuracy	MCC
Useless model (always A)	95.0% ← looks great!	0.000 ← correctly says "random"
Good model	92.0%	0.750

Accuracy would **wrongly** rank the useless model higher! MCC correctly identifies it as no better than chance.

MCC Worked Example (Binary)

$TP = 40,\ TN = 45,\ FP = 5,\ FN = 10$

$$MCC = \frac{40 \times 45 - 5 \times 10}{\sqrt{(40+5)(40+10)(45+5)(45+10)}} = \frac{1800 - 50}{\sqrt{45 \times 50 \times 50 \times 55}} = \frac{1750}{\sqrt{6{,}187{,}500}} = \frac{1750}{2487.5} = \mathbf{0.}$$

Interpretation: Strong positive correlation — the model's predictions align well with truth.

Cohen's Kappa (κ)

The Core Question

What Cohen's Kappa Asks

"How much better is the model than pure random chance?"

Kappa adjusts accuracy by subtracting out the agreement that would happen purely by luck. A model that gets 90% accuracy on an easy problem (where random guessing gets 80%) is not as impressive as one that gets 80% on a hard problem (where random guessing gets 2%).

The Formula

Mathematical Definition

$$\boxed{\kappa = \frac{p_o - p_e}{1 - p_e}}$$

where:

- p_o = **observed agreement** = accuracy = fraction of correct predictions
- p_e = **expected agreement by chance** = probability that a random model would be correct, given the class distribution

For multi-class:

$$p_e = \sum_{c=1}^{C} \frac{n_{c,\text{pred}} \times n_{c,\text{true}}}{N^2}$$

where $n_{c,\text{pred}}$ = how many times the model predicted class c, and $n_{c,\text{true}}$ = how many times class c actually appeared.

Worked Example

Cohen's Kappa Step-by-Step

Given: 100 test samples, 2 classes (A and B). The confusion matrix is:

	Predicted A	Predicted B	Total (True)
True A	40	10	50
True B	5	45	50
Total (Pred)	45	55	100

Step 1: Observed agreement: $p_o = \frac{40+45}{100} = 0.85$

Step 2: Expected agreement by chance:

$$p_e = \frac{45 \times 50}{100^2} + \frac{55 \times 50}{100^2} = \frac{2250}{10000} + \frac{2750}{10000} = 0.225 + 0.275 = 0.50$$

(A random model would get 50% right just by the distribution of predictions and labels.)

Step 3: Kappa:

$$\kappa = \frac{0.85 - 0.50}{1 - 0.50} = \frac{0.35}{0.50} = \mathbf{0.70}$$

Standard Interpretation Scale

κ Range	Agreement Level	Your Results
< 0.00	Less than chance	
$0.01 - 0.20$	Slight	
$0.21 - 0.40$	Fair	
$0.41 - 0.60$	Moderate	
$0.61 - 0.80$	Substantial	Qwen HMDB51 ($\kappa = 0.765$)
$0.81 - 1.00$	**Almost Perfect**	**V-JEPA 2: 0.974 (FG), 0.842 (HM)**

Kappa vs. MCC: What's the Difference?

	MCC	Cohen's Kappa (κ)
Origin	Statistics (Pearson's ϕ)	Psychology (inter-rater reliability)
Measures	Correlation between predictions & truth	Improvement over random chance
Range	$[-1, +1]$	$[-1, +1]$ (typically $[0, 1]$)
In practice	Often $\approx \kappa$	Often $\approx$ MCC

In your results, MCC and κ are nearly identical (e.g., both 0.974 on FineGym, both ≈ 0.843 on HMDB51). Reporting **both** gives independent confirmation from two different statistical traditions.

The Confusion Matrix (Foundation of All Metrics)

What It Is

The Confusion Matrix

A $C \times C$ table (where C = number of classes) that shows **exactly how many samples of each true class were predicted as each class**.

Row i, Column j = number of samples whose true label is i but were predicted as j.

Visual Example: FineGym (Near-Perfect Model)

True Label \ Predicted Label	BB	FX	UB	VT	
BB	**73**	0	1	0	← 73 out of 74 BB correct
FX	0	**63**	0	0	
UB	0	0	**78**	1	← 1 UB misclassified as VT
VT	0	0	0	**56**	

How to Read the Confusion Matrix

1. **Diagonal cells** (green) = **correct predictions**. Larger numbers = better.
2. **Off-diagonal cells** (errors) show *which classes get confused with which.* For example, if cell (UB, BB) = 5, it means 5 Uneven Bars videos were wrongly classified as Balance Beam.
3. **Row sums** = total actual samples per class.
4. **Column sums** = total predictions per class.

What the Confusion Matrix Reveals That No Single Number Can

The confusion matrix answers questions like:

- "Which two classes does the model confuse most often?" (look for large off-diagonal values)
- "Is there one class the model almost never gets right?" (look for a row with

small diagonal value)

- "Does the model over-predict a particular class?" (column sum much larger than row sum)

No single metric (accuracy, F1, MCC, etc.) can tell you WHERE the errors are — only the confusion matrix can.

Why We Need ALL These Metrics Together

No single metric tells the complete story. Each one catches different failure modes.

What Each Metric Catches That Others Miss

Metric	What It Reveals	What It Misses
Top-1 Accuracy	How often the best guess is correct	Whether rare classes are ignored; whether the model is "close" when wrong
Top-5 Accuracy	Whether the model "understands" the problem even when wrong	Doesn't distinguish #2 from #5; inapplicable for few classes
Precision	How trustworthy positive predictions are	Whether the model finds all instances (recall)
Recall	How completely the model finds each class	Whether the "found" predictions are correct (precision)
F1-Score	Balance of precision and recall	Doesn't account for true negatives; can be gamed
MCC	Overall correlation using ALL confusion matrix cells	Doesn't show class-specific problems
Cohen's κ	Improvement over random chance	Similar to MCC; adds interpretive scale

The Story They Tell Together

Using your V-JEPA 2 HAF-STD Net results on HMDB51 as an example:

Complete Metric Story: V-JEPA 2 on HMDB51

Metric	Value	What This Tells Us
Top-1 Accuracy	84.7%	The model gets the exact right answer 85% of the time
Top-5 Accuracy	95.8%	When wrong, the right answer is usually in the top 5
Precision (macro)	87.7%	When it makes a prediction, it's usually correct
Recall (macro)	84.8%	It finds most instances of each action class
F1 (macro)	85.3%	Precision and recall are well-balanced
MCC	0.843	Strong genuine correlation, not a class-imbalance artifact
Cohen's κ	0.842	Almost perfect agreement, far better than chance

Combined narrative: The model is highly accurate (84.7%), and even when it's wrong, it's usually "close" (95.8% top-5). Its predictions are trustworthy (87.7% precision) and comprehensive (84.8% recall), with good balance between the two (85.3% F1). This performance is genuine, not an artifact of class distribution (MCC = 0.843), and represents almost perfect agreement beyond chance ($\kappa = 0.842$).

Danger of Using Only One Metric

What Goes Wrong With Single Metrics

If you only report...	A reviewer could ask...
Accuracy alone	"Is this inflated by class imbalance? What about minority classes?"
F1 alone	"Is the model confident? How does it rank against chance?"
Top-1 alone	"Does the model completely fail when wrong, or is it close?"
MCC alone	"What's the actual accuracy? Which specific classes fail?"

Reporting all seven metrics preemptively answers every possible reviewer question about your model's classification quality.

Quick Reference Card

All Metrics at a Glance

Metric	Range	Perfect	Random	Your Best (V-JEPA 2)	
				FineGym	HMDB51
Top-1 Accuracy	$[0, 1]$	1.0	$1/C$	98.1%	84.7%
Top-5 Accuracy	$[0, 1]$	1.0	$5/C$	N/A	95.8%
Precision (macro)	$[0, 1]$	1.0	$\sim 1/C$	98.5%	87.7%
Recall (macro)	$[0, 1]$	1.0	$\sim 1/C$	98.3%	84.8%
F1 (macro)	$[0, 1]$	1.0	$\sim 1/C$	97.8%	85.3%
MCC	$[-1, +1]$	$+1.0$	0.0	0.974	0.843
Cohen's κ	$[-1, +1]$	$+1.0$	0.0	0.974	0.842

Where C = number of classes (FineGym: $C = 4$, HMDB51: $C = 51$).

Key takeaway: For FineGym ($C = 4$), random chance gives 25% accuracy. Your model achieves 98.1%. For HMDB51 ($C = 51$), random chance gives ~2% accuracy. Your model achieves 84.7%. Both are **overwhelmingly better than chance**, confirmed by MCC and κ values close to 1.0.

Remember: Accuracy tells you "how often." F1 tells you "how balanced." MCC tells you "how genuine." Kappa tells you "how much better than luck." Top-5 tells you "how close when wrong." Together, they tell the **full story**.